WAIS

Web Agent Interaction Standard

Specification Draft v0.1

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DRAFT — For Review and Feedback

An open standard for enabling AI agents to interact with, transact on, and complete actions on websites on behalf of authenticated human users.

1. Executive Summary

The Web Agent Interaction Standard (WAIS) defines how websites can expose transactional capabilities to AI agents acting on behalf of authenticated human users. While existing standards like llms.txt focus on content discoverability and readability, WAIS addresses the next frontier: enabling agents to perform actions—purchasing products, submitting forms, managing returns, booking services, and completing multi-step workflows—in a structured, secure, and verifiable way.

WAIS is built on a core principle we call Proof of Delegation: an agent does not need to prove it is human. It needs to prove that a real human has delegated it the authority to act.

This document presents the initial specification for community review and feedback. The goal is to establish an open, vendor-neutral standard that benefits websites, agent platforms, and end users equally.

2. The Problem

2.1 The Current State

Today, when an AI agent needs to perform an action on a website on behalf of a user, it must resort to brittle, unreliable methods: headless browser automation, DOM scraping, visual element detection, and heuristic-based form filling. These approaches break frequently, cannot handle CAPTCHAs, and are indistinguishable from malicious bots.

Meanwhile, websites invest heavily in blocking automated access, creating an adversarial dynamic that hurts both sides: agents cannot reliably serve users, and websites lose potential customers whose agents cannot complete transactions.

2.2 The Paradigm Shift

The internet is transitioning from a human-first interface to one where AI agents are primary intermediaries. Users increasingly want to delegate tasks—“buy this list of items,” “return this product,” “book me a flight”—to agents that act on their behalf. Websites that make this easy will capture this traffic. Those that don’t will become invisible to an entire channel of commerce.

WAIS provides the bridge: a structured way for websites to welcome verified agents and for agents to prove they act with legitimate human authorization.

2.3 What Exists Today

Several standards address parts of this problem, but none covers the full interaction lifecycle:

Standard	Focus	Gap
llms.txt	Content discoverability for LLMs	Read-only; no actions, no auth

robots.txt	Crawler access permissions	Binary allow/block; no interaction
Schema.org	Structured data markup	Descriptive, not actionable
OAuth 2.0	User authorization for apps	Designed for apps, not agents
OpenAPI	API specification	Assumes dedicated API; most sites lack one

WAIS fills the gap by defining how agents discover, authenticate, and execute actions on any website, whether it has an API or not.

3. Core Principles

- **Proof of Delegation over Proof of Humanity.** Agents don't prove they're human. They prove a human authorized them.
- **Progressive Capability.** Sites can adopt WAIS incrementally—from a simple manifest to full transactional support.
- **Human in the Loop.** High-risk actions always require explicit user confirmation. The standard defines when and how.
- **Open and Vendor-Neutral.** No single agent platform or website technology should be required.
- **Security by Default.** All agent interactions must be authenticated, scoped, and auditable.
- **Privacy Preserving.** Sites can verify agent authorization without knowing the user's identity.

4. Architecture Overview

4.1 The Three Parties

Every WAIS interaction involves three entities:

- **The User** — A real human who wants to accomplish a task.
- **The Agent** — An AI system acting on behalf of the user (e.g., Claude, ChatGPT, a custom agent).
- **The Site** — A website or web service that offers actions the agent can perform.

4.2 Interaction Flow

The standard interaction flow follows five phases:

#	Phase	Description
1	**Discovery**	Agent fetches the site's agents.json manifest to learn what actions are available.
2	**Authentication**	Agent presents its Proof of Delegation token, identifying itself, its

		platform, and the user's authorization scope.
3	**Execution**	Agent invokes declared actions using the schemas defined in the manifest.
4	**Confirmation**	For actions above the defined risk threshold, the site returns a confirmation challenge that the agent must present to the user for approval.
5	**Reporting**	Site returns structured results that the agent can interpret and present to the user.

5. The Manifest: agents.json

5.1 Overview

The agents.json file is the entry point for WAIS. Placed at the root of a website (e.g., https://example.com/agents.json), it declares what an agent can do on the site, how to authenticate, and what confirmation requirements exist for each action.

5.2 Structure

A minimal agents.json file:

```
{
  "wais_version": "0.1",
  "site": {
    "name": "ExampleStore",
    "description": "Online electronics retailer",
    "support_contact": "agents@example.com"
  },
  "authentication": {
    "methods": ["wais-delegation-token"],
    "token_endpoint": "/wais/auth/verify"
  },
  "actions": [
    {
      "id": "search_products",
      "description": "Search product catalog",
      "endpoint": "/wais/api/products/search",
      "method": "POST",
      "risk_level": "low",
      "requires_confirmation": false,
      "input_schema": { ... },
```

```
      "output_schema": { ... }
    },
    {
      "id": "checkout",
      "description": "Complete purchase",
      "endpoint": "/wais/api/checkout",
      "method": "POST",
      "risk_level": "high",
      "requires_confirmation": true,
      "confirmation_details": ["total_amount", "items",
        "shipping_address", "estimated_delivery"],
      "input_schema": { ... },
      "output_schema": { ... }
    }
  ]
}
```

5.3 Action Risk Levels

Level	Description	Examples
low	Read-only or reversible actions with no financial impact	Search products, view order status, browse catalog
medium	Reversible actions or low financial commitment	Add to cart, save to wishlist, update profile, initiate return
high	Irreversible actions or significant financial commitment	Complete purchase, cancel subscription, submit legal forms
critical	High-value actions requiring strong user authentication	Large transactions, contract signing, delete account

6. Proof of Delegation Protocol

6.1 Concept

Proof of Delegation (PoD) is the authentication mechanism at the heart of WAIS. It inverts the traditional web security model: instead of proving a visitor is human, PoD proves that an automated agent has been explicitly authorized by a real human to act within defined boundaries.

6.2 The Delegation Token

A Delegation Token is a cryptographically signed credential that encodes:

- **Agent Identity:** Who is this agent? A verifiable identifier tied to the agent platform (e.g., Anthropic, OpenAI, a registered independent agent).
- **User Authorization:** Proof that a real human consented to this delegation. The user's identity can remain anonymous to the site—only the fact of authorization needs verification.
- **Scope:** What actions is this agent authorized to perform? Scopes are granular and follow a standard taxonomy (e.g., catalog.browse, cart.modify, checkout.execute, return.initiate).
- **Constraints:** Limits on the delegation (e.g., maximum transaction amount, time validity, geographic restrictions).
- **Platform Signature:** Cryptographic signature from the agent's platform, verifiable via public key infrastructure.

6.3 Token Structure (Simplified)

The delegation token uses a JWT-like structure:

```
{
  "header": {
    "alg": "ES256",
    "typ": "WAIS-PoD",
    "kid": "anthropic-agent-key-2026-02"
  },
  "payload": {
    "iss": "https://anthropic.com",
    "sub": "agent:claude-session-abc123",
    "aud": "https://example-store.com",
    "iat": 1708617600,
    "exp": 1708621200,
    "delegation": {
      "user_hash": "sha256:a1b2c3...",
      "user_verified": true,
      "consent_timestamp": 1708617500,
      "scopes": [
        "catalog.browse",
        "cart.modify",
        "checkout.execute"
      ],
      "constraints": {
        "max_transaction_amount": {
          "value": 500,
          "currency": "EUR"
        },
```

```
        "require_confirmation_above": {
          "value": 100,
          "currency": "EUR"
        }
      }
    }
  },
  "signature": "base64url-encoded-signature"
}
```

6.4 Verification Flow

When an agent presents a PoD token to a site:

1. The site validates the token signature against the agent platform's published public key.
2. The site checks that the token hasn't expired and that the audience matches its domain.
3. The site verifies that the requested action falls within the token's declared scopes.
4. If the action exceeds constraints (e.g., amount over threshold), the site returns a confirmation challenge.
5. No CAPTCHAs. No browser fingerprinting. Trust is established cryptographically.

6.5 Distinguishing Agents from Bots

A key benefit of PoD is that it cleanly separates legitimate agents from malicious bots:

Property	WAIS Agent	Malicious Bot
Delegation token	Valid, signed by known platform	Absent or forged
Scoped permissions	Granular, declared in advance	Attempts everything
User behind it	Verified human (hashed)	No user linkage
Confirmation flow	Supports challenge-response	Cannot complete
Audit trail	Full traceability	None

7. Confirmation Protocol

7.1 When Confirmation Is Required

The site defines confirmation requirements per action in agents.json. WAIS recommends the following defaults, which sites can customize:

- **Low risk:** No confirmation required. Agent can execute freely.

- **Medium risk:** Soft confirmation. Agent must present action details to user before proceeding.
- **High risk:** Hard confirmation. Site generates a challenge; user must approve via their agent platform.
- **Critical risk:** Strong authentication. User must authenticate directly (biometric, PIN, 2FA) through their agent platform.

7.2 Confirmation Challenge Structure

When a site requires confirmation, it responds to the agent with a structured challenge:

```
{
  "wais_confirmation": {
    "challenge_id": "conf_abc123",
    "action": "checkout",
    "risk_level": "high",
    "expires_at": "2026-02-22T15:30:00Z",
    "display_to_user": {
      "summary": "Purchase 3 items from ExampleStore",
      "total": "€247.50",
      "items": [
        "Sony WH-1000XM5 Headphones - €189.00",
        "USB-C Cable 2-pack - €12.50",
        "Screen Protector - €46.00"
      ],
      "shipping": "Standard (3-5 business days)",
      "payment_method": "Visa ending 4242"
    },
    "approval_methods": ["user_confirm", "biometric"]
  }
}
```

The agent’s platform presents this to the user in its own interface, collects approval, and returns a signed confirmation response to the site.

8. Standard Scopes by Vertical

WAIS defines a taxonomy of scopes to ensure consistency across implementations. Sites declare which scopes they support; agents request only what they need.

8.1 E-Commerce

Scope	Allows	Risk Level
catalog.browse	Search and view products	Low
catalog.compare	Access pricing and availability for comparison	Low
cart.modify	Add/remove items from cart	Medium
checkout.execute	Complete a purchase	High
order.track	View order status and tracking	Low
return.initiate	Start a return or exchange	Medium
return.complete	Complete return process (label, pickup)	High
subscription.manage	Modify or cancel subscriptions	High

8.2 Travel & Hospitality

Scope	Allows	Risk Level
availability.search	Search flights, hotels, restaurants	Low
booking.create	Make a reservation	High
booking.modify	Change dates, seats, rooms	High
booking.cancel	Cancel a reservation	High
claim.submit	File compensation claims	Medium

8.3 Financial Services

Scope	Allows	Risk Level
account.read	View balances and transactions	Medium
quote.request	Request insurance/loan quotes	Low
payment.execute	Make a payment or transfer	Critical
dispute.file	Dispute a charge	Medium
policy.modify	Change insurance policy details	High

8.4 Government & Public Services

Scope	Allows	Risk Level

appointment.book	Schedule appointments	Medium
form.submit	Submit government forms	High
document.request	Request documents or certificates	High
status.check	Check application or process status	Low

8.5 Healthcare

Scope	Allows	Risk Level
appointment.schedule	Book medical appointments	Medium
prescription.renew	Request prescription renewals	High
records.access	Retrieve test results	Critical

9. Use Cases in Detail

9.1 Agent-Powered Shopping

A user tells their agent: ***"Prepare a list of supplies for my camping trip and buy everything for under €300."***

The agent searches multiple WAIS-compliant outdoor stores simultaneously (catalog.browse), compares prices and availability (catalog.compare), builds an optimized cart across the best-priced stores, presents a consolidated summary to the user: "Found everything across 2 stores for €267. Confirm?", and upon user approval, executes checkout at each store (checkout.execute with confirmation protocol).

Post-purchase, the agent monitors order status (order.track) and if any item arrives damaged, initiates a return (return.initiate) and schedules a pickup (return.complete)—all without the user visiting a single website.

9.2 Travel Disruption Management

A user's flight gets cancelled. Their agent automatically:

Detects the cancellation via the airline's status endpoint, searches alternative flights (availability.search) across multiple airlines, books the best option within the user's preferences (booking.create with confirmation), adjusts the hotel reservation dates (booking.modify), and files an EU261 compensation claim (claim.submit)—all within minutes, presenting each decision to the user for approval.

9.3 Subscription Audit

*A user asks: **"Review all my subscriptions and cancel anything I haven't used in 3 months."***

The agent accesses each subscription service (account.read), checks usage data, presents a report to the user, and upon approval, cancels the unused ones (subscription.manage with confirmation for each). Dark patterns that hide cancellation flows become irrelevant because the agent interacts via the structured WAIS endpoint, not the UI.

9.4 Government Bureaucracy

A user needs to renew their driver's license. The agent:

Checks eligibility and requirements (status.check), fills out the renewal form with the user's stored data (form.submit with critical confirmation), uploads required documents, books an appointment if in-person visit is needed (appointment.book), and tracks the application status—turning a multi-hour bureaucratic process into a 2-minute interaction.

10. Adoption Path

10.1 Progressive Implementation

WAIS is designed for incremental adoption. Sites can start minimal and expand:

Level 1: Manifest Only

Publish agents.json with read-only actions (search, browse, status). Zero backend changes required if the site already has any kind of API. This alone makes the site discoverable to agents.

Level 2: Basic Interactions

Add medium-risk actions (add to cart, update preferences). Implement PoD token verification. The site now supports assisted workflows where agents prepare actions for users.

Level 3: Full Transactions

Enable high-risk actions (checkout, returns, cancellations) with the confirmation protocol. This is the full WAIS experience where agents can complete end-to-end workflows.

Level 4: Proactive Integration

Implement webhooks and event streams so agents can react to changes (order shipped, price drop, booking cancelled). The site becomes a fully agent-native platform.

10.2 Incentive Structure

The adoption incentive is competitive pressure, not regulation. The first e-commerce site in a category to implement WAIS captures agent-driven traffic that competitors cannot serve. As more users delegate shopping to agents, non-WAIS sites lose a growing channel.

This mirrors the early days of e-commerce: the first retailers to accept online payments captured demand that brick-and-mortar stores could not. WAIS is the “accept online payments” moment for the agent era.

11. Security Considerations

- **Token Expiration:** All PoD tokens must have a short TTL (recommended: 1 hour max). Long-lived delegations should use refresh mechanisms.
- **Scope Minimization:** Agents should request the minimum scopes needed. Sites should reject over-scoped tokens.
- **Replay Protection:** Tokens should include a nonce and be bound to a specific session.
- **Rate Limiting:** Sites should implement per-agent rate limits separate from human rate limits.
- **Audit Logging:** All agent actions must be logged with the agent’s identity and delegation context for accountability.
- **Revocation:** Users must be able to revoke agent permissions at any time via their agent platform. Sites should support token revocation checks.
- **Platform Registry:** A public registry of trusted agent platforms and their public keys enables sites to verify agent identities.

12. What Comes Next

12.1 Immediate Next Steps

1. Community review and feedback on this draft specification.
2. Reference implementation of agents.json validator and PoD token library.
3. Proof-of-concept integration with one e-commerce platform.
4. Formal JSON Schema definitions for agents.json and PoD tokens.
5. Open-source audit tool that scans sites for WAIS compliance.

12.2 Open Questions

Several design decisions require community input:

- Should the Platform Registry be centralized (like a CA system) or decentralized (like DNS)?
- How should WAIS interact with existing authentication flows (OAuth, SAML, passkeys)?
- What governance model should manage the standard’s evolution?
- How should WAIS handle sites that require user accounts vs. guest checkout?
- Should there be a WAIS compliance certification program?

13. Get Involved

WAIS is an open initiative. We are seeking feedback from web developers, e-commerce platforms, AI agent builders, security researchers, and anyone who believes the future of the web involves agents acting on behalf of humans.

If your website wants to be where agents shop, book, file, and transact in the agent era, WAIS is how you open that door.

The future isn't about proving you're human. It's about proving a human trusts you.